

DAY 25: MULTIPLE LINEAR REGRESSION

The Absolute OLS Matrix Formula Proof & Clean Production Code Blueprint

Welcome to Day 25. Today, we thoroughly dissect the exact analytical core of Multiple Linear Regression. No random updates, no loops—just pure linear algebra that unlocks global minimums instantly.

Day 25 me aapka swagat hai. Aaj hum Multiple Linear Regression ke exact analytical core ka poora post-mortem karenge. Na koi andaza, na koi loop—sirf pure linear algebra jo instantly global minimum error nikaal kar deta hai.

1. THE 3-DAY LEDGER MATRIX DATASET

Our standard production-level data vectors mapped for structural matrix operations:

Matrix operations ke liye hamara standard production-level data vectors niche diye gaye hain:

- **Day 1:** Weight (X_1) = 10, Complexity (X_2) = 3 → Price (Y) = 80,000
- **Day 2:** Weight (X_1) = 30, Complexity (X_2) = 4 → Price (Y) = 3,75,000
- **Day 3:** Weight (X_1) = 20, Complexity (X_2) = 5 → Price (Y) = 2,50,000

2. DEEP DIVE: THE OLS MATRIX FORMULA PROOF

When multiple features are present, writing separate scalar derivatives for every slope becomes highly inefficient. We condense everything into a clean matrix system.

Jab features ek se zyada ho jaate hain, toh har ek slope ke liye alag se scalar derivative likhna mushkil ho jata hai. Isliye hum poore system ko matrix form me convert karte hain.

The Matrix Setup

Let our feature matrix be X (with an added column of 1s to perfectly handle the baseline intercept c), our target column vector be Y , and our parameter weight vector be B :

Maan lijiye hamara feature matrix X hai (jisme 1s ka ek column pehle se joda jata hai taaki intercept c handle ho sake), target vector Y hai, aur saare weights/parameters ka vector B hai:

$$Y = XB + \epsilon$$

Where ϵ represents the residual error vector. Our target is to minimize the Sum of Squared Residuals (SSR):

Yahan ϵ hamara bacha hua error (residual) vector hai. Hamara target hota hai Residuals ke squares ke sum (SSR) ko sabse kam karna:

$$SSR = \epsilon^T \epsilon = (Y - XB)^T (Y - XB)$$

Expanding the Error Equation

By applying matrix multiplication distribution properties, we fully expand the transposes:

Matrix multiplication ke niyam ke mutabik transpose ko open karke equation ko expand karte hain:

$$SSR = Y^T Y - B^T X^T Y - Y^T X B + B^T X^T X B$$

Since the term $B^T X^T Y$ is a scalar, it is exactly equal to its own transpose $Y^T X B$. Thus, we can combine them perfectly:

Kyunki $B^T X^T Y$ ek scalar value hai, yeh apne hi transpose $Y^T X B$ ke barabar hoti hai. Isliye hum in dono ko aapas me jod sakte hain:

$$SSR = Y^T Y - 2B^T X^T Y + B^T X^T X B$$

Applying Matrix Calculus (Taking the Derivative)

To find the absolute lowest error point, we take the matrix derivative with respect to B and set it exactly to 0:

Sabse lowest error point dhondhne ke liye, hum is poori equation ka derivative B ke respect me nikaalte hain aur use 0 ke barabar rakh dete hain:

$$\partial(SSR) / \partial B = -2X^T Y + 2X^T X B = 0$$

Now, shifting terms to balance the balance sheet matrix:

Ab terms ko shift karke matrix ka balance sheet barabar karte hain:

$$2X^T X B = 2X^T Y$$

$$X^T X B = X^T Y$$

To isolate our optimal weight vector B , we multiply both sides by the matrix inverse $(X^T X)^{-1}$. This gives us the legendary, globally definitive OLS formula:

Optimal weights vector B ko akela karne ke liye, hum dono taraf $(X^T X)^{-1}$ ke matrix inverse se multiply karte hain. Isse hamara legendary aur universal OLS formula nikal kar aata hai:

$$\mathbf{B} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$$

Why it's flawless: This single equation matrix calculation does not require a single loop or iterations. It analytically arrives at the 100% correct answer on the very first try.

Yeh flawless kyun hai: Is single matrix calculation ko chalane ke liye ek bhi loop ya step-by-step andaze ki zarurat nahi parti. Yeh pehli hi baar me 100% sateek answer analytical tareeke se laakar rakh deta hai.

3. CLEAN PRODUCTION CODE BLUEPRINT

Below is the highly structured, clean, and professional python code designed using standard practices:

Niche ekdum saaf, clear aur professional layout me python code likha gaya hai standard syntax ke sath:

```

import numpy as np
from sklearn.linear_model import LinearRegression

# =====
# 1. CORE DATASET SETUP (Multi-Dimensional Feature Arrays)
# Columns represent structural inputs: [Gold Weight, Design Complexity]
# =====
X_train = np.array([
    [10, 3], # Day 1 Input Vector
    [30, 4], # Day 2 Input Vector
    [20, 5] # Day 3 Input Vector
])

# Actual Target Market Prices vector
y_true = np.array([80000, 375000, 250000])

# =====
# 2. MODEL INITIALIZATION & ANALYTICAL OLS TRAINING
# =====
regressor = LinearRegression()
regressor.fit(X_train, y_true)

# =====
# 3. EXTRACTING METICULOUS MATHEMATICAL PARAMETERS
# =====
weight_m1 = regressor.coef_[0] # Optimized Weight for Feature X1
weight_m2 = regressor.coef_[1] # Optimized Weight for Feature X2
intercept_c = regressor.intercept_ # Globally Perfect Base Intercept

# =====
# 4. EXECUTIVE OUTPUT DISPLAY
# =====
print("--- PRODUCTION OPTIMIZED ANALYTICAL OUTPUT ---")
print(f"Optimized m1 (Gold Weight Coefficient) : {weight_m1:.2f}")
print(f"Optimized m2 (Design Complexity Slope) : {weight_m2:.2f}")
print(f"Optimized c (Showroom Baseline Intercept): {intercept_c:.2f}")

```

4. FINAL ANALYTICAL SUMMARY

By learning the matrix proof of OLS, you have uncovered how machine learning algorithms optimize parameters instantly. When your clean code executes this, scikit-learn runs the identical matrix transposition and inversion proved above to give you the most accurate answer possible.

OLS ke matrix proof ko samajh kar aapne yeh jaan liya hai ki machine learning backend par parameters ko instantly kaise nikaalti hai. Jab aapka yeh clean code execute hoga, toh scikit-learn parde ke peeche isi matrix inversion aur transposition ka use karke aapko poori duniya ka sabse sateek answer laakar dega.